size_balanced

equity strategy | family: equity-baseline | status: validated | derived from `extreme_picks`

What this strategy does

Extreme Picks, with an equal budget for each of the five company sizes

Why it should work. PEAD is stronger in small caps; balancing by size keeps the edge but removes a size bet.

Design notes (from the strategy's source file).

Size Balanced -- Extreme Picks with an equal budget for each company-size group.

Extreme Picks' trade, but each leg's budget is spread evenly over the size quintiles present that quarter, so micro-caps (where most extreme surprises happen) cannot dominate the book.

parameter	value
base_unit_fraction	0.003
max_gross_leverage	1.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	proportional
exposure_cap	open_book
beta_overlay	0.0

Performance

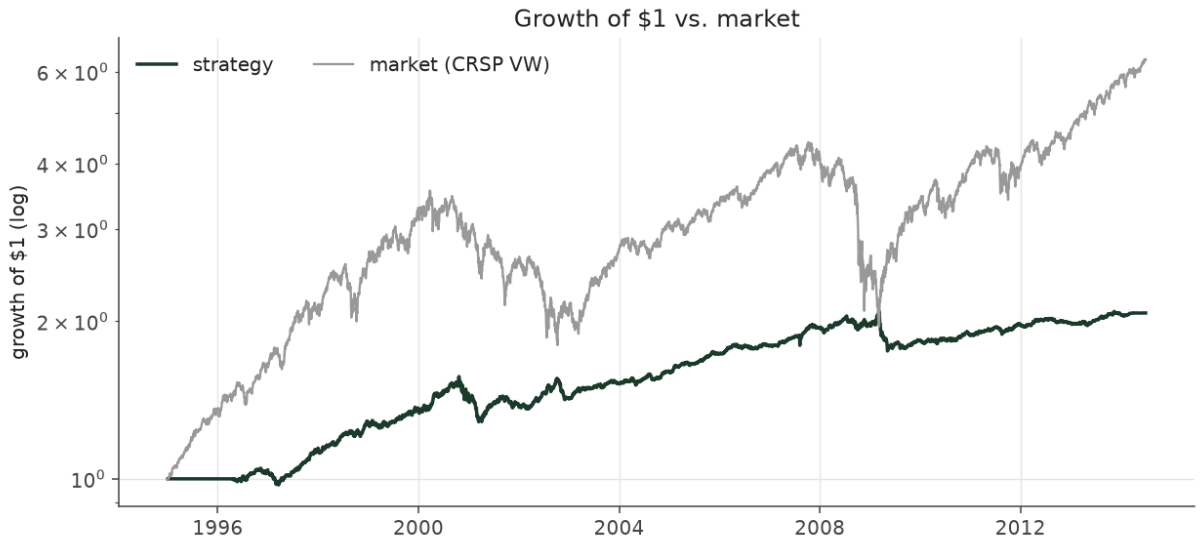
metric	size balanced	market (CRSP VW)
Annualized return	3.82%	9.94%
Annualized volatility	4.77%	19.35%
Sharpe (NAV-pct)	0.81	0.59
Sortino	1.13	
Max drawdown	-18.1%	-55.5%
Calmar	0.21	
Final NAV ($10M start)	$20,749,649	
Worst day	-2.56%	
% positive days	50.7%	
Sharpe 1996-2001	0.93	
Sharpe 2002-2007	1.44	
Sharpe 2008-2013	0.24	

Correlation of daily returns with the market: **0.01**.

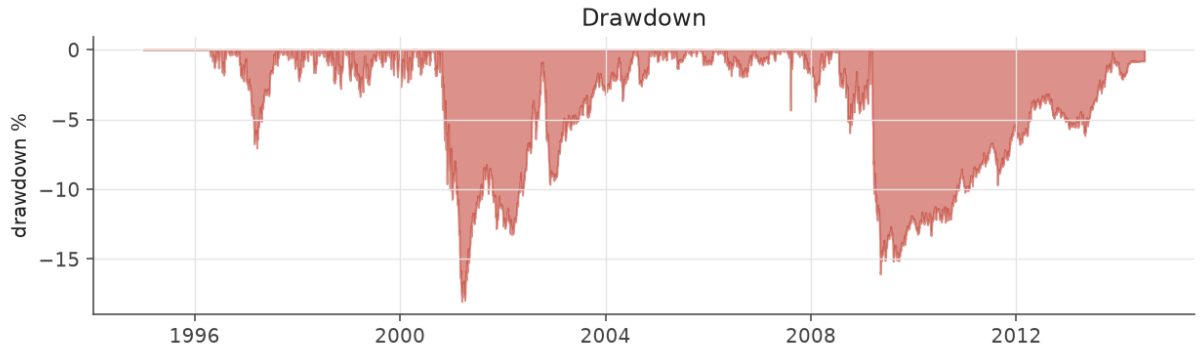
• best Sharpe among equity: `calm_rank_tilt` (1.81); this one: 0.81

• best return among equity: `market_boost_max` (0.15); this one: 0.0382

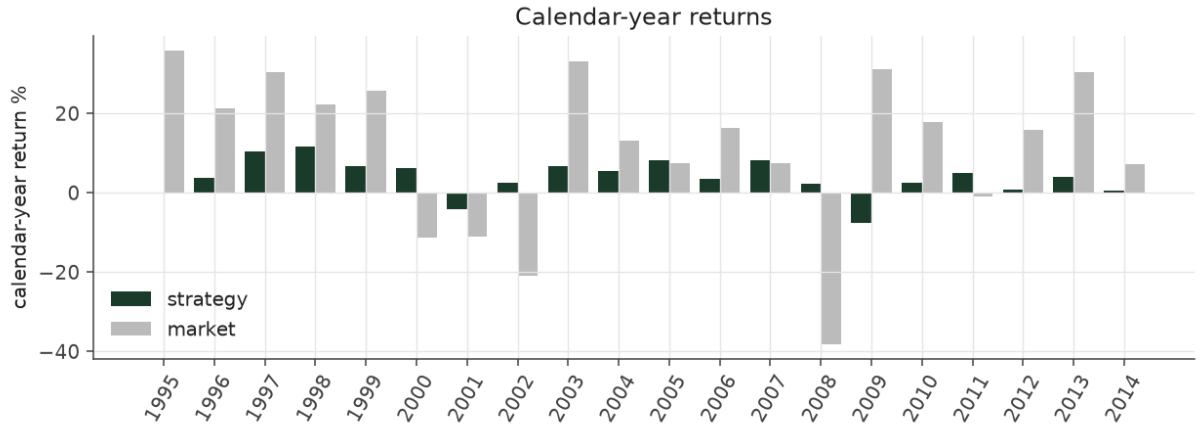
• best Calmar among equity: `calm_rank_tilt` (0.614); this one: 0.212



NAV (log scale)



Drawdown from running peak

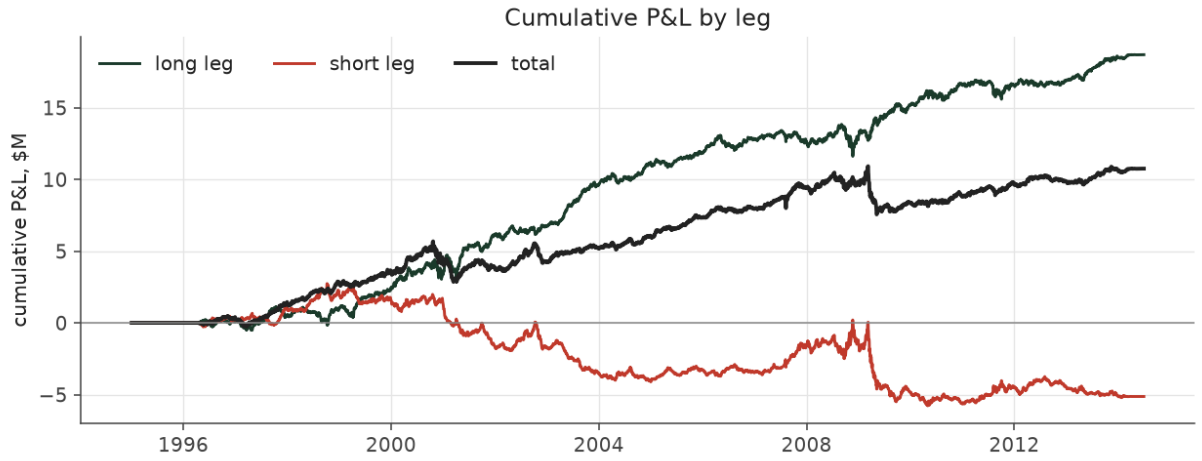


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	18.69	-5.14
P&L after costs ($M)	17.36	-6.61
of which common (average-event) component ($M)	10.73	-11.29
of which excess / PEAD component ($M)	7.96	6.15
excess minus costs ($M)	6.64	4.67
gross return per $ traded (bp)	276.40	-72.18
positions in the money (hit rate)	0.57	0.49
stand-alone Sharpe (leg only)	1.09	-0.27

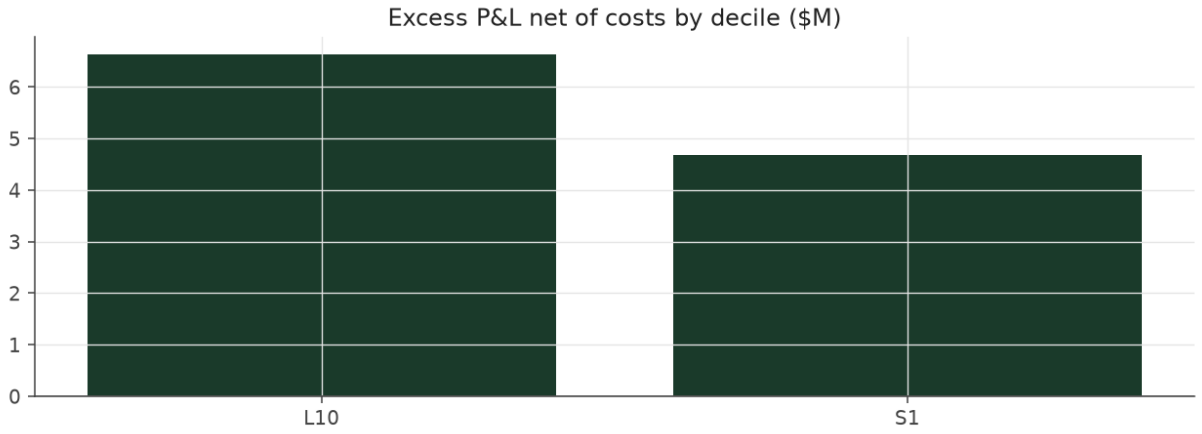
Daily P&L correlation between the two legs: **-0.66**.

• **The short leg loses money in absolute terms** (-5.1M before costs) while the long leg earns +18.7M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 10.7M / 11.3M. Stripping that common component, the surprise signal itself earned +8.0M on the long side and +6.1M on the short side.

• The two legs' daily P&L are -0.66 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.09 long / -0.27 short vs 0.81 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	10	20,136	676.22	17.36	7.96	1.33	6.63	58.7%	98.12
short	1	18,299	711.41	-6.61	6.15	1.48	4.67	41.3%	65.69



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	5,287	137.15	-0.50	0.55	0.50	0.05	0.5%	3.85
2	7,872	290.58	4.00	4.90	0.80	4.10	36.2%	141.04
3	8,605	315.31	3.94	4.50	0.65	3.85	34.1%	122.22
4	8,744	321.55	1.92	2.42	0.50	1.92	16.9%	59.55
5	7,927	323.05	1.39	1.75	0.36	1.39	12.3%	42.99

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	4,793	86.25	1.63	2.37	0.43	1.94	17.2%	225.26
2	7,905	297.87	1.45	2.82	0.89	1.93	17.1%	64.75
3	8,745	341.48	4.54	5.37	0.68	4.69	41.5%	137.36
4	8,837	339.62	1.21	1.71	0.47	1.23	10.9%	36.30
5	8,153	322.31	1.94	1.86	0.32	1.53	13.6%	47.64

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	7,482	325.98	0.95	5.04	0.61	4.42	39.1%	135.71
41-60d	11,345	387.22	-0.23	1.45	0.86	0.58	5.2%	15.05

61-90d	17,258	600.32	6.54	4.93	1.13	3.80	33.6%	63.34
>90d	2,350	74.11	3.49	2.70	0.20	2.50	22.1%	337.12

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	10,157	351.52	3.80	4.59	0.67	3.92	34.7%	111.48
2002-07	13,612	479.65	5.38	6.39	0.96	5.44	48.1%	113.34
2008-13	14,666	556.46	1.57	3.13	1.18	1.95	17.3%	35.09

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	7,541	268.24	1.30	0.68	0.54	0.14	1.3%	5.29
Chems	776	27.88	0.02	0.12	0.05	0.07	0.6%	24.42
Durbl	990	34.25	0.16	0.37	0.07	0.29	2.6%	86.04
Enrgy	1,529	58.75	1.23	1.43	0.12	1.31	11.6%	223.64
Hlth	3,443	120.86	0.93	0.73	0.26	0.47	4.2%	39.03
Manuf	4,089	144.11	0.33	1.08	0.29	0.79	7.0%	55.08
Money	5,981	219.86	1.37	2.95	0.44	2.52	22.2%	114.41
NoDur	1,707	60.78	0.46	0.55	0.12	0.43	3.8%	70.21
Other	5,674	202.06	2.27	2.72	0.45	2.27	20.1%	112.41
Shops	3,935	140.88	2.19	2.54	0.28	2.25	19.9%	160.03
Telcm	1,293	50.55	0.54	0.76	0.10	0.67	5.9%	132.17
Utils	1,477	59.41	-0.05	0.19	0.10	0.09	0.8%	14.83

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	12,942	467.67	0.46	1.60	0.85	0.75	6.6%	16.00
low vol	12,792	453.85	5.46	6.73	1.01	5.71	50.5%	125.92
mid vol	12,052	436.63	4.26	5.22	0.88	4.34	38.4%	99.37

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	3.82%	0.00%
0.5%	0.7	3.63%	0.19%
1.0%	1.5	3.43%	0.39%
2.0%	2.9	3.01%	0.81%
5.0%	7.3	1.52%	2.30%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.